

Application Note

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Abstract

Over the years, sample preservation has been used to maintain sample integrity by reducing the rate of microbiological growth, which can cause sample contamination or degradation. Acid preservation has become the method of choice for Total Organic Carbon (TOC) analysis for many official methods. The addition of inorganic acid to the sample stops the action of metabolizing the organics for food by the microorganisms present in the sample. The decreased pH of the sample also retards the growth of slime molds that are present as spores in the atmosphere. However, there has become a debate over which acid is best to use to preserve the sample without interfering with the sample analysis. Comparison data will be examined defining the pros and cons of acid preservation on two widely used TOC technologies, persulfate and combustion.

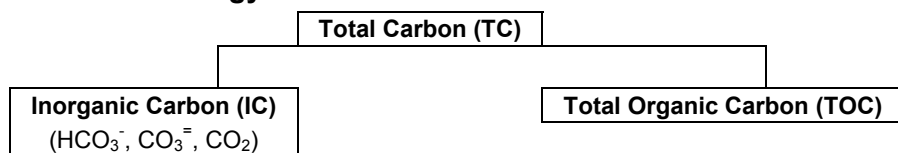
Background

Almost all TOC methodologies that prescribe sample acidification call for the pH to be lowered to ≤ 2 pH at the sampling time. However, methods vary in the type of acid recommended for their preservation requirements, as seen in Table 1.

Method	Last Update	Acid Preservation Recommendation
EPA 415.1 ¹ (Combustion or Oxidation)	1974	pH ≤ 2 with HCl or H ₂ SO ₄ within 2 hours of sampling
EPA 415.2 ² Test Method (UV promoted, persulfate oxidation)	1982	pH ≤ 2 with H ₂ SO ₄ within 2 hours of sampling <i>Special Warning:</i> HCl should not be used.
ASTM D 2579-85 ³	1985	pH between 2 & 3
EPA 9060A ⁴	1990	pH ≤ 2 with HCl or H ₂ SO ₄ within 2 hours of sampling
ASTM D 4779-93 ⁵	1993	pH = 2 +/-1 with H ₃ PO ₄
Information Collection Rule ⁶	1994	Refer to Standard Methods 5310 C
ASTM D 4839-94 ⁷	1994	pH = 2 with H ₃ PO ₄ , HNO ₃ or H ₂ SO ₄
Standard Methods 5310 B & C ⁸	1996	pH ≤ 2 with H ₃ PO ₄ or H ₂ SO ₄
ISO/ FDIS 8245 ⁹	1998	pH = 2 (H ₃ PO ₄ is used as an example.)
USP 24 –NF 19 Chapter<643> ¹⁰	2000	No Recommendations on Acid Preservation. Only “test in a timely manner to minimize organic contamination”.

Table 1. History of TOC preservation

TOC Methodology



TOC analysis by sparging is known as Non-Purgeable Organic Carbon (NPOC) analysis. This is the most commonly used method for TOC analysis because of its high throughput, excellent precision, and great accuracy on high IC containing samples. TOC by the NPOC method involves acidification of a sample followed by sparging of that sample with nitrogen, zero grade (hydrocarbon-free) air or oxygen. This process is used to remove the Inorganic Carbon, which is typically vented to the atmosphere in the form of CO_2 .

IC Removal

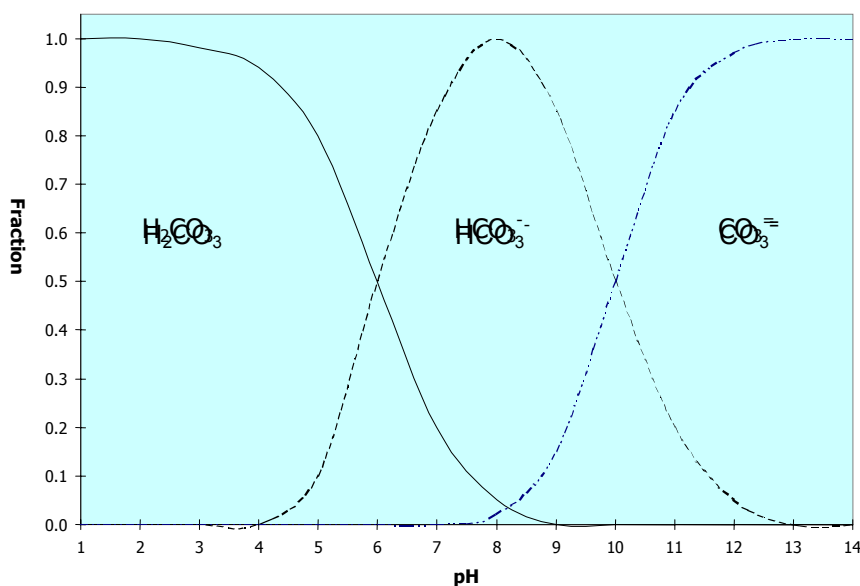
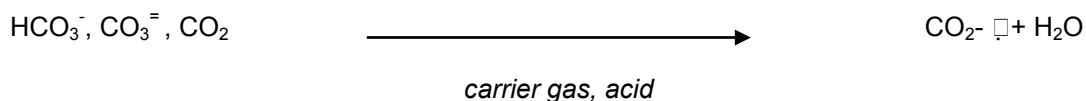


Figure 1: CO_2 Distribution in Water versus pH

As the pH is lowered through acidification, carbonates and bicarbonates form carbonic acid, H_2CO_3 , as seen in figure 1.¹¹ Carbonic acid is also known as dissolved CO_2 and H_2O . The dissolved carbon dioxide portion can be removed by sparging the solution with a carrier gas such as nitrogen, zero grade (hydrocarbon-free) air, or oxygen.



Experimental

Since many present day TOC analyzers will automatically add an aliquot of acid to samples before sparging to remove IC, problems can arise from adding acid to the already acidified samples collected in the field. In the case of pre-acidified samples, additional acid should not be used.

In addition to over acidified samples, one should also consider the possible effects of acids on the TOC analyzer. The effects of the most commonly used acids in preservation, such as hydrochloric, sulfuric, nitric and phosphoric, were studied to determine the effects on two TOC analyzers for tap water samples. The two technologies used were high temperature catalytic combustion oxidation analyzer, the Apollo 9000, and a UV promoted, persulfate oxidation analyzer, the Phoenix 8000. Both techniques were followed by Non-Dispersive Infrared (NDIR) detection for TOC.

Results

Phoenix 8000 – UV/Persulfate TOC Analyzer		
Sample*	Result (n = 3 repetitions)	% RSD
Tap Water acidified with H ₃ PO ₄	0.8480 ppmC	0.79
Tap Water preserved in H ₂ SO ₄	0.9050 ppmC	1.01
Tap Water preserved in HNO ₃	0.8483 ppmC	2.03
Tap Water preserved in HCl	0.6587 ppmC	18.53

Apollo 9000 - Catalytic Combustion TOC Analyzer		
Sample*	Result (n = 3 repetitions)	% RSD
Tap Water acidified with H ₃ PO ₄	0.8601 ppmC	1.70
Tap Water preserved in H ₂ SO ₄	1.2866 ppmC	3.05
Tap Water preserved in HNO ₃	0.9114 ppmC	3.80
Tap Water preserved in HCl	0.8598 ppmC	2.25

* Each tap water sample was acidified to 2 pH. This typically required 2 – 3 drops of the aforementioned acids.

Phosphoric Acid (H₃PO₄)

A weaker acid compared to all other acids mentioned.¹² However, it is not as corrosive and the combustion product is P₂O₅, which is readily taken up in water and does not stay in a gaseous state that would be swept into the detector.

The results above demonstrate the performance of both TOC technologies under the recommended automated acid addition of phosphoric acid.

Nitric Acid (HNO₃)

Nitric acid will form nitric oxides in different quantities for the two techniques; more for combustion and less for persulfate. In combustion instruments, N₂O₄ will be formed. The halide scrubber will remove some of the gas but not all, causing the potential for corrosion in the detector.

The results indicate that nitric acid preservation yields similar performance to phosphoric acid on the UV persulfate technique. Whereas the combustion technique shows a 6% higher result with higher % RSD compared to the phosphoric acid.

Hydrochloric Acid (HCl)

HCl will form HCl gas for both combustion and persulfate techniques. Since most commercial analyzers use a gold-lined sample cell in the NDIR detector to detect CO₂, the corrosive nature of HCl gas can damage the sample cell of the NDIR.^{12, 13} Most TOC analyzers are equipped with halide scrubbers that remove most of the HCl when maintained properly.

For persulfate TOC analyzers, another problem exist with the nature of HCl preserved samples, as *Standard Methods 5310C* states:

“Persulfate oxidation of organics molecules is slowed in the samples containing significant concentrations of chlorides by the preferential oxidation of chloride; at concentrations above 0.05% chloride, oxidation of organic matter may be inhibited.”⁸

This effect can be seen in the low results for HCl preserved samples run on the Phoenix 8000 with high % RSD compared to phosphoric acid.

On the Apollo 9000, the catalytic combustion TOC analyzer, similar results were obtained for HCl preserved samples as for phosphoric acid preserved samples. Through the combustion process, the HCl gas passes through the combustion tube and catalyst unaffected the oxidation process. Then, the HCl gas enters a halide scrubber that allows the effectively remove the HCl gas. Hence, with the proper maintenance of the halide scrubber, the HCl preserved samples are similar to phosphoric acid preserved tap water samples.

Sulfuric Acid (H₂SO₄)

Sulfuric acid will form SO₃ gas in both combustion and persulfate TOC instruments. Since SO₃ has similar detection properties as CO₂, the SO₃ will cause a positive interference in the NDIR detector. Hence, a positive interference can be observed on both TOC analyzers for sulfuric acid-preserved samples in the form of higher results compared to phosphoric acid-preserved samples.

A SO₃ scrubbing device can be used ahead of the halide scrubber that will remove the SO₃ from the gas that will enter the detector without removing CO₂. Tekmar-Dohrmann recommends installing a SO₃ mist scrubber when running samples that contain sulfuric acid on both the Phoenix 8000 and Apollo 9000 analyzers.

Conclusion

Testing TOC samples in a timely manner is still the best method to prevent organic contamination and degradation. Analysis done in this manner is also the best way to prevent the possible effects from the acid or the combinations of acids used. However, in most labs, the volume of samples, time needed to transport samples or method recommendations dictate the use of acid preservation. When choosing a preservation acid, one should study the effects of acids on their instrumental methodology.

For UV-Persulfate Oxidation with NDIR detection, phosphoric acid is the acid of choice for its less corrosive properties. Nitric acid would be a good alternative for this methodology. Sulfuric acid could be used with the addition of a SO₃ scrubber. While hydrochloric acid should be avoided.

For catalytic combustion oxidation with NDIR detection, phosphoric acid is the acid of choice for its less corrosive properties. Hydrochloric acid would be a good alternative for this methodology. However, the halide scrubber would need to be changed more frequently. Nitric and sulfuric acid would be a final alternative. Sulfuric acid would require the use of a SO₃ scrubber.

Acid Used to Preserve Sample	Persulfate Oxidation	Catalytic Combustion
H ₃ PO ₄	Use with no concern of interference	Use with no concern of interference
HCl	Not Recommended	Use as an alternative to H ₃ PO ₄ ; Maintain Corrosive Scrubber
HNO ₃	Use as an alternative to H ₃ PO ₄ ; Maintain Corrosive Scrubber	Not Recommended
H ₂ SO ₄	Not Recommended	Not Recommended

Reference

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